

## Answers to AP Precalc Unit 1 Wkst D: Unit Review

1. Q1
2. Q3
3. Q3
4. Q1
5. Q4
6. Q2
7. Q1
8. Q1
9.  $78^\circ$  and  $-282^\circ$
10.  $-\frac{\pi}{2}, \frac{3\pi}{2}$
11.  $-\frac{5\pi}{6}, \frac{19\pi}{6}$
12.  $-234^\circ, 486^\circ$
13.  $255^\circ, -465^\circ$
14.  $-\frac{17\pi}{12}, \frac{31\pi}{12}$
15.  $\frac{5\pi}{4}, -\frac{11\pi}{4}$
16.  $288^\circ, -432^\circ$
17. a)  $\frac{\pi}{3}$  b)  $\frac{5\pi}{6}$
18. a) None b)  $\frac{\pi}{4}$
19. a)  $\frac{\pi}{12}$  b)  $\frac{7\pi}{12}$
20. a) None b) None
21. a) None b) None
22. a) None b)  $\frac{\pi}{5}$
23.  $\frac{3\pi}{4}$
24.  $120^\circ$
25.  $225^\circ$
26.  $\frac{3\pi}{2}$
27.  $\frac{4\pi}{3}$
28.  $30^\circ$
29.  $330^\circ$
30.  $\frac{11\pi}{6}$
33.  $\frac{\sqrt{3}}{2}$
34.  $\frac{\sqrt{3}}{3}$
35.  $-\frac{\sqrt{2}}{2}$
36.  $-1$
37.  $1$
38.  $\frac{1}{2}$
39.  $-2$
40.  $-1$
- $\sin \theta = \frac{4}{5} \quad \cot \theta = \frac{3}{4}$
41.  $\cos \theta = \frac{3}{5} \quad \sec \theta = \frac{5}{3}$   
 $\tan \theta = \frac{4}{3} \quad \csc \theta = \frac{5}{4}$
- $\sin \theta = -\frac{2}{\sqrt{13}} \quad \cot \theta = -\frac{3}{2}$
42.  $\cos \theta = \frac{3}{\sqrt{13}} \quad \sec \theta = \frac{\sqrt{13}}{3}$   
 $\tan \theta = -\frac{2}{3} \quad \csc \theta = -\frac{\sqrt{13}}{2}$
43.  $-\frac{\sqrt{11}}{6}$
44.  $-\frac{5}{13}$
45.  $\frac{\sqrt{33}}{4}$
46.  $-\frac{\sqrt{5}}{2}$
47.  $-5\sqrt{2}$  (square root of 50 breaks  
down to 5 root 2)
48.  $-\frac{\sqrt{6}}{6}$  or  $-\frac{1}{\sqrt{6}}$
49.  $\frac{\sqrt{26}}{5}$
50.  $-\frac{1}{\sqrt{5}}$  or  $-\frac{\sqrt{5}}{5}$
51.  $\frac{\pi}{6}, \frac{5\pi}{6}$

52.  $\frac{5\pi}{6}, \frac{7\pi}{6}$
53.  $\frac{3\pi}{4}, \frac{7\pi}{4}$
54.  $\frac{2\pi}{3}, \frac{5\pi}{3}$
55.  $\frac{\pi}{3}, \frac{5\pi}{3}$
56.  $\frac{7\pi}{6}, \frac{11\pi}{6}$
57.  $\frac{\pi}{3}, \frac{4\pi}{3}$
58.  $\frac{\pi}{4}, \frac{3\pi}{4}$
59.  $0, \pi, 2\pi$
60.  $\pi$
61.  $0, \pi, 2\pi$
62.  $\frac{3\pi}{4}, \frac{5\pi}{4}$
63.  $\frac{\pi}{6}$
64.  $55^\circ$
65.  $87^\circ$
66. 0.42
67.  $\frac{\pi}{3}$
68. 0.64
69. 0.2588
70. -1.2983
71. 0.0709
72. -0.7071
73. 0.4663
74. 17.1309
75.  $14.99^\circ, 165.01^\circ$
76.  $105.54^\circ, 254.46^\circ$
77.  $81.57^\circ, 261.57^\circ$
78.  $146.31^\circ, 326.31^\circ$
79.  $41.81^\circ, 138.19^\circ$
80.  $346.19^\circ, 193.81^\circ$
81.  $89.76^\circ, 270.24^\circ$
82.  $83.66^\circ, 276.34^\circ$
83.  $\frac{5}{11}$  radians
84. 2 radians
85.  $18\pi$  meters
86.  $3\pi$  feet
87.  $\frac{164}{3\pi}$  feet
88.  $\frac{384}{7\pi}$  feet
89. a)  $400\pi$  radians  
b) The distance travelled  
each minute is  $2380\pi$  cm  
c)  $\frac{119\pi}{3}$  in/sec
90. The distance travelled is  
 $1560\pi$  cm in each minute.
91.  $AS = 4\pi$  rad/sec;  
 $LS = 8\pi$  in/sec
92.  $AS = 10\pi$  rad/sec  
 $LS = 120\pi$  in/sec  
 $LS \approx 21.42$  mph